

SOROCHINSK, Russian Federation

WMO: 350110

Lat: 52.433N

Lon: 53.133E

Elev: 403

StdP: 14.48

Time Zone: 5.00 (E05)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.2	-12.6	-24.0	1.5	-18.0	-18.0	2.1	-11.8	19.0	22.7	17.8	22.0	2.1	120	0.543

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.6	94.4	66.2	90.9	65.2	87.8	64.5	68.7	85.7	67.4	84.3	66.2	83.2	6.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.0	90.5	72.9	62.1	84.6	72.5	60.3	79.2	71.8	33.1	85.7	32.0	83.7	31.0	83.6	73.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.1	15.7	13.8	-24.4	98.3	6.4	2.1	-29.0	99.8	-32.7	101.0	-36.3	102.2	-40.9	103.7		
(5)			-24.6	70.7	6.3	1.2	-29.2	71.6	-32.9	72.3	-36.4	72.9	-41.0	73.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	42.6	9.3	11.4	24.0	45.5	60.8	68.7	72.3	69.7	57.7	42.6	29.7	17.3	(6)
(7)		DBStd	24.37	11.72	11.51	9.41	10.03	7.74	8.20	6.03	8.57	7.77	8.52	9.29	11.37	(7)
(8)		HDD50	5267	1261	1080	807	205	8	1	0	1	21	259	610	1014	(8)
(9)		HDD65	8852	1726	1500	1272	587	177	54	11	51	240	695	1060	1479	(9)
(10)		CDD50	2559	0	0	0	70	343	561	692	613	252	28	0	0	(10)
(11)		CDD65	670	0	0	0	3	48	164	237	198	20	0	0	0	(11)
(12)		CDH74	7961	0	0	0	45	641	2013	2756	2269	237	0	0	0	(12)
(13)		CDH80	3304	0	0	0	3	163	886	1159	1042	51	0	0	0	(13)
(14)	Wind	WSAvg	6.2	6.2	5.8	6.9	7.0	6.2	6.0	5.4	5.4	5.6	6.6	6.7	6.5	(14)
(15)	Precipitation	PrecAvg	16.1	1.2	0.9	1.1	1.1	1.4	2.0	1.5	1.3	1.4	1.4	1.3		(15)
(16)		PrecMax	23.2	2.2	1.9	2.8	3.5	3.7	5.6	3.6	3.8	3.6	3.1	3.6	3.0	(16)
(17)		PrecMin	11.6	0.2	0.2	0.1	0.2	0.1	0.0	0.0	0.2	0.1	0.1	0.2	0.2	(17)
(18)		PrecStd	3.0	0.5	0.4	0.7	0.8	0.9	1.3	1.1	0.9	0.9	0.7	1.0	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.1	34.5	51.1	79.7	87.8	96.0	97.8	98.2	85.9	69.1	51.0	37.8	(19)
(20)			MCWB	31.8	33.1	43.7	57.7	61.4	65.9	67.8	66.3	60.4	52.3	47.3	36.6	(20)
(21)		2%	DB	31.3	32.7	40.1	73.7	84.5	92.9	93.9	94.0	80.6	63.5	47.1	35.2	(21)
(22)			MCWB	30.4	31.6	36.0	55.3	60.7	65.1	67.5	65.7	59.4	50.5	43.9	33.9	(22)
(23)		5%	DB	28.9	30.7	37.4	67.7	81.2	89.5	89.9	90.2	76.1	59.2	44.0	33.7	(23)
(24)			MCWB	28.1	29.4	34.3	52.6	59.8	64.0	66.0	64.9	57.6	48.5	41.4	32.5	(24)
(25)		10%	DB	25.8	27.6	35.3	62.2	77.4	85.4	86.5	86.4	71.9	55.1	41.2	32.1	(25)
(26)			MCWB	24.7	26.1	32.9	49.5	58.2	63.7	64.9	64.2	56.5	47.1	39.4	30.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.3	33.6	43.8	59.4	63.8	69.0	71.1	69.2	63.9	55.8	48.4	36.9	(27)
(28)			MCD B	33.0	34.2	50.6	75.5	80.7	87.6	88.9	85.2	79.5	63.0	50.2	37.5	(28)
(29)		2%	WB	30.5	31.8	36.6	56.6	62.3	67.5	69.6	67.7	61.4	52.1	44.3	34.2	(29)
(30)			MCD B	31.3	32.6	39.2	70.8	79.2	85.3	87.0	84.9	73.8	59.5	46.4	34.8	(30)
(31)		5%	WB	28.1	29.6	34.7	53.7	60.9	66.2	68.0	66.6	59.8	50.1	41.8	32.8	(31)
(32)			MCD B	28.8	30.7	36.8	64.9	78.3	83.3	84.2	84.8	71.2	55.9	43.5	33.6	(32)
(33)		10%	WB	24.7	26.1	33.2	51.0	59.2	65.0	66.6	65.5	58.0	48.3	39.5	30.9	(33)
(34)			MCD B	25.8	27.5	35.1	61.5	74.4	81.0	82.4	83.0	68.8	53.9	41.2	32.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.7	13.6	13.4	18.0	21.7	22.3	22.6	21.7	18.6	14.0	8.5	9.9	(35)
(36)			MCDBR	10.3	9.5	12.3	26.2	27.3	27.5	27.1	26.4	26.4	21.5	9.5	7.2	(36)
(37)			MCWBR	10.4	8.9	9.3	12.0	10.0	9.1	8.8	8.0	10.7	11.3	7.5	7.3	(37)
(38)		5% WB	MCD BR	10.4	8.3	11.4	22.9	25.0	22.9	23.0	23.7	21.3	17.1	8.7	7.2	(38)
(39)			MCWBR	10.5	8.1	9.2	11.6	10.3	8.6	8.5	8.1	9.8	10.3	7.8	7.4	(39)
(40)																
(41)	Clear-Sky Solar Irradiance	taub	0.302	0.327	0.377	0.433	0.409	0.402	0.409	0.428	0.382	0.351	0.330	0.302	(41)	
(42)		taud	2.057	2.020	1.963	2.063	2.250	2.326	2.306	2.234	2.324	2.319	2.223	2.077	(42)	
(43)		Ebn at Noon	205	238	249	252	267	270	266	252	251	233	198	181	(43)	
		Edh at Noon	25	35	45	47	41	38	38	39	32	26	21	20		
(44)	All-Sky Solar Radiation	RadAvg	342	703	1150	1435	1817	1984	1935	1613	1095	563	252	209	(44)	
(45)		RadStd	47	51	130	155	144	203	166	135	147	97	43	50	(45)	

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		N/A	N/A	N/A	+1.80	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page